

Sprint 1 Bootstrap: 36-State Factored MDP with LLM-Estimated Transitions

William Dennis

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1 MDP Formulation

The Sprint 1 simulator implements a finite-horizon factored MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$ with 36 states, 4 actions, and transitions estimated by an LLM (deepseek-v4-flash) using Algorithm 2.

1.1 State Space

Four factors: step bin (`step_bin` $\in \{\text{inactive}, \text{moderate}, \text{active}\}$), sleep quality (`sleep` $\in \{\text{good}, \text{poor}\}$), day type (`day_of_week` $\in \{\text{weekday}, \text{weekend}\}$), and burden level (`burden` $\in \{\text{low}, \text{medium}, \text{high}\}$):

$$\mathcal{S} = \{\text{inactive}, \text{moderate}, \text{active}\} \times \{\text{good}, \text{poor}\} \times \{\text{weekday}, \text{weekend}\} \times \{\text{low}, \text{medium}, \text{high}\}$$

Three factors are deterministic: `day_of_week` cycles daily through a 7-day pattern; `burden` uses a rolling window of the last 3 actions (increments on non-idle actions). Two factors are stochastic: `step_bin` at each timestep, and `sleep` at day boundaries.

1.2 Action Space

$$\mathcal{A} = \{\text{idle}, \text{movement_suggestion}, \text{goal_reminder}, \text{journal}\}$$

All non-idle actions incur a penalty of 0.05.

1.3 Transition Function

Transitions are loaded from 6 JSON probability tables generated by an LLM (deepseek-v4-flash, $N = 16$ samples per state-action pair, Algorithm 2):

- `day_boundary.json`: $P(\text{sleep}' \mid \text{step_bin}, \text{burden}, \text{day_of_week}, \text{sleep})$ – action-independent, evaluated at step 0 of each day.
- `within_day_0.json` through `within_day_4.json`: $P(\text{step_bin}' \mid \text{step_bin}, \text{burden}, \text{action}, \text{day_of_week}, \text{sleep})$ – one table per timestep, action-conditioned.

The LLM’s estimates reflect plausible real-world dynamics: interventions have modest effects on step counts, and sleep quality influences next-day activity.

1.4 Reward

$$R = \alpha \cdot \text{step_bin_value} + (1 - \alpha) \cdot \text{sleep_value} - \text{action_penalty}$$

with $\alpha = 0.9$. Step bin values: `inactive` $\rightarrow$ 0.0, `moderate` $\rightarrow$ 0.5, `active` $\rightarrow$ 1.0. Sleep values: `good` $\rightarrow$ 1.0, `poor` $\rightarrow$ -1.0. Action penalty: 0.0 for idle, 0.05 otherwise.

Optional `reward_multiplier_by_step` masks the last step of each day: $[1, 1, 1, 1, 0]$.

1.5 Episode

$T = 90 \text{ days} \times 5 \text{ steps/day} = 450 \text{ timesteps per episode}$.

2 Reference Policy Bounds

Six reference policies evaluated via simulation (10 seeds, 450 timesteps):

Policy	Total Reward	Per Step	Description
Fixed idle	380.4 ± 3.7	0.845	Never intervene (no penalty)
Greedy oracle (1-step)	281.0 ± 20.9	0.625	Myopic expected-reward maximisation
Random	100.1 ± 11.7	0.222	Uniform action selection
Fixed movement_suggestion	35.2 ± 4.8	0.078	Always suggest movement
Fixed journal	50.5 ± 5.1	0.112	Always journal
Fixed goal_reminder	91.2 ± 7.1	0.203	Always remind of goal

Table 1: Reference policy bounds (10 seeds). Always-idle dominates because the LLM estimates natural activity is already high and action penalties are costly. The greedy oracle’s myopic one-step lookahead fails to capture long-term value.

Always-idle at 0.845/step sets an upper bound for agents: the LLM estimates that most users maintain moderate-to-active step counts without intervention, and paying a 0.05 penalty for no improvement is worse than doing nothing.

3 Base Benchmark Results

50 seeds, 450 timesteps, default hyperparameters (EG $\epsilon = 0.1$, UCB $c = 2.0$, TS $\alpha_0 = \beta_0 = 1$). Config: `configs/sprint1_bootstrap.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Epsilon-Greedy ($\epsilon = 0.1$)	280.8 ± 89.3	0.624	0.743
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	252.6 ± 105.6	0.561	0.662
UCB ($c = 2.0$)	240.4 ± 33.9	0.534	0.803
Random	99.7 ± 10.9	0.222	0.215

Table 2: Base benchmark results (50 seeds, 450 timesteps). Learning agents massively outperform random (2.5–2.8 $\times$), confirming the bootstrap tables contain exploitable structure. EG leads.

Key observation: under bootstrap transitions, learning agents achieve 2.5–2.8 $\times$ random, starkly different from the random-transition baseline where all agents clustered within 3% of each other. The LLM-estimated dynamics have meaningful structure.

4 Full Benchmark Results

50 seeds, 450 timesteps, 9 agents with MVP-optimised hyperparameters (EG $\epsilon = 0.05$, D-EG $\epsilon_{\text{start}} = 0.2$, $T_{\text{decay}} = 200$, UCB $c = 0.5$). Config: `configs/sprint1_bootstrap_extensions.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	358.1 ± 14.0	0.796	0.837
Standard D-EG	287.8 ± 106.9	0.640	0.718
Standard EG ($\epsilon = 0.05$)	282.3 ± 95.0	0.627	0.748
Contextual UCB	268.3 ± 29.6	0.596	0.733
Standard TS	252.6 ± 105.6	0.561	0.662
Contextual D-EG	240.4 ± 93.0	0.534	0.614
Contextual EG	216.9 ± 95.7	0.482	0.583
Contextual TS	155.5 ± 70.8	0.346	0.399
Random	99.7 ± 10.9	0.222	0.215
Fixed idle (upper bound)	380.4	0.845	0.845

Table 3: Full benchmark results (50 seeds). Standard UCB at $c = 0.5$ reaches 94% of the always-idle ceiling. Contextual agents universally underperform their standard counterparts.

Key findings:

1. **Std UCB at $c = 0.5$ is dominant:** 358.1 total, 94% of the always-idle ceiling, with the lowest variance (14.0 vs 70–106 for others). Hyperparameter tuning is critical: the default $c = 2.0$ yielded only 240.4.
2. **Contextual agents are worse:** Every contextual variant underperforms its standard counterpart, the opposite of the MVP result. This suggests `step_bin` as a context feature misleads agents in the bootstrap dynamics – transitions are non-stationary (vary by timestep) and the current `step_bin` does not reliably predict intervention efficacy.
3. **The $c = 0.5$ UCB achieves consistency:** with ± 14.0 std dev vs ± 33.9 at $c = 2.0$, tuning simultaneously improves both mean and variance.

5 Masked Reward Benchmark

Same 9 agents on `configs/sprint1_bootstrap_extensions_masked.yaml` where step 4 (end of day) yields zero reward. Results are 50 seeds.

6 Comparison to Random-Transition Baseline

7 Context Feature Comparison

The extensions benchmark used `step_bin` as the context feature (3 bins, same fragmentation as `step_bin`). To test whether a better-chosen context could close the gap, we swapped to `burden` (3 levels, same fragmentation) and re-ran the 9-agent benchmark. Results are 50 seeds.

Despite having $13\times$ higher ANOVA F-statistic as a predictor of next-step activity, `burden` makes a worse context feature. The reason is that `burden` is *endogenous* – it is a rolling window count

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	265.9 ± 37.3	0.591	0.681
Standard D-EG	248.3 ± 76.2	0.552	0.616
Standard EG ($\epsilon = 0.05$)	195.8 ± 98.4	0.435	0.518
Contextual UCB	178.4 ± 35.2	0.397	0.501
Standard TS	174.6 ± 70.6	0.388	0.456
Contextual D-EG	169.9 ± 70.0	0.378	0.435
Contextual EG	163.7 ± 73.9	0.364	0.443
Contextual TS	119.0 ± 38.5	0.264	0.289
Random	83.6 ± 8.4	0.186	0.179

Table 4: Masked benchmark results (50 seeds, step 4 zero-reward). Relative ordering is preserved. D-EG is most mask-resilient (86% retention).

Metric	Random Transitions	Bootstrap Transitions
Best agent	Std D-EG 189.1 (0.420)	Std UCB 358.1 (0.796)
Random baseline	183.5 (0.408)	99.7 (0.222)
Best vs random gap	3%	259%
Contextual advantage	None ($\approx 0\%$)	Negative (all worse)
Variance (best agent)	± 82	± 14

Table 5: Bootstrap transitions vs random transitions. The LLM-estimated tables produce structured dynamics with large agent differentials, while random transitions wash out all policy differences. Std UCB’s low variance under bootstrap suggests reliable convergence.

of the agent’s past non-idle actions. When a fragmented contextual agent explores, it takes non-idle actions, which increase burden, which shifts the context distribution, which triggers further exploration. This creates a feedback loop:

exploration $\rightarrow$ higher burden $\rightarrow$ unfamiliar context $\rightarrow$ more exploration $\rightarrow \dots$

In contrast, `step_bin` is at least partly exogenous – it depends on the LLM-estimated transition dynamics and is not directly controlled by recent actions. The `step_bin` distribution is relatively stable regardless of the agent’s policy, making it a safer (if less predictive) context feature.

Both context features are dominated by the standard (non-contextual) UCB, confirming that `step_bin` context does not help this MDP.

8 Key Findings

1. **Bootstrap transitions have exploitable structure.** Unlike the Dirichlet-random baselines where all agents performed similarly (~ 185 total), the LLM-estimated tables yield a 259% gap between best learner and random. The LLM produces non-trivial transition dynamics.
2. **Always-idle is near-optimal.** The LLM estimates that most users naturally maintain moderate-to-active step counts. Interventions incur a penalty without consistently improving

Agent	No Context	Ctx <code>step_bin</code>	Ctx <code>burden</code>
Standard UCB ($c = 0.5$)	358.1 ± 14.0	—	—
Contextual UCB	—	268.3 ± 29.6	250.6 ± 39.7
Contextual D-EG	—	240.4 ± 93.0	216.0 ± 77.6
Contextual EG	—	216.9 ± 95.7	200.4 ± 82.9
Contextual TS	—	155.5 ± 70.8	158.3 ± 56.1
Random	99.7 ± 10.9	—	—

Table 6: Comparison of context features (50 seeds). Burden context underperforms `step_bin` context across all agents except TS (where the difference is negligible).

outcomes, making inaction the best policy. Std UCB learns this implicitly, reaching 94% of the idle ceiling.

3. **Context features hurt, even with better signal.** Swapping from `step_bin` to `burden` – a context with $13\times$ stronger predictive power – made performance worse, not better. The reason is that `burden` is endogenous: exploration changes `burden`, which shifts the context distribution, which triggers more exploration. This feedback loop amplifies the sample-efficiency cost of context fragmentation.
4. **UCB hyperparameter sensitivity is extreme.** Default $c = 2.0$: 240. Tuned $c = 0.5$: 358. The MVP-optimised values transferred well, but a dedicated grid search may find even better settings.
5. **The greedy oracle (one-step lookahead) is not a good bound** at 281, far below Std UCB’s 358. Myopic expected-reward maximisation fails because actions affect future states through `burden` accumulation and sleep at day boundaries.
6. **The LLM’s transition estimates are a plausible starting point** but should be validated against real data. The strong idle preference may reflect LLM bias toward inaction rather than true user dynamics.